I. **TITLE OF THE PROPOSAL:** Proposed Offering of **Master of Information Technology – Professional Track** in the College of Engineering and Computational Science at Partido State University for AY 2026-2027 through the adoption of CHED Memorandum Order (CMO) No. 15, s.2019

II. **PROPONENT(S):** College of Engineering and Computational Sciences, Bachelor of Science in Information Technology (Level IV Accredited)

III. **OVERVIEW/RATIONALE**

The world is currently experiencing an unprecedented process of transformation, which Klaus Schwab of the World Economic Forum called the 4th Industrial Revolution. In his book "The 4th Industrial Revolution," he declared that new technologies fuse the physical, digital, and biological worlds around us. In another book by Nancy Gleason entitled "Higher Education in the Era of the 4th Industrial Revolution," she asserted that we are now in a new phase where the fusion of several technologies is not only automating production but also knowledge. The changes are occurring because humans have finally developed the computing capacity to store massive amounts of data, which in turn can enable machine learning. The outcome of this is the development of what are called Cyber-Physical Systems (CPS). Ragunathan Rajkumar explained that CPS are physical and engineered systems whose operations are monitored, coordinated, controlled, and integrated by a computing and communication core. Just as the internet transformed how humans interact with one another, CPS will transform how we interact with the physical world around us. Gleason added that many grand challenges await in the economically vital domains of transportation, health care, manufacturing, agriculture, energy, defense, aerospace, and buildings.

With all this said, expertise in information technology will place students in the mainstream of the 4th industrial revolution. In whatever aspect of the life of our society, graduates of MIT will be the most sought-after group.

According to the Elsevier Research Group, the speed of current breakthroughs has no historical precedent. When compared with previous industrial revolutions, the fourth is evolving at an exponential rather than a linear pace. Moreover, it is disrupting almost every industry in every country. And the breadth and depth of these changes herald the transformation of entire systems of production, management, and governance. These possibilities will be multiplied by emerging technology breakthroughs in fields such as artificial intelligence, automation, robotics, the Internet of Things, autonomous vehicles, 3-D printing, nanotechnology, biotechnology, materials science, energy storage, and quantum computing.

The proposed Master of Information Technology (MIT) program is designed to align with the global standards and frameworks established under ISO 21001:2018 for educational management systems, ensuring consistency in meeting learner's needs and fostering sustainable development in accordance with the UN SDGs, emphasizing quality education, inclusive opportunities for all learners, and sustainable cities and communities. The MIT program is designed to contribute meaningfully to the UNs Sustainable Development Goals (SDGs) by equipping students with the knowledge, skills, and innovative mindset needed to address global challenges through technology. The program aligns with SDG 4: Quality Education by providing advanced, industry-relevant education that empowers students with technical and vocational skills essential for employment, entrepreneurship, and lifelong learning in the digital economy. By fostering adaptability and critical thinking, the program ensures graduates are prepared to thrive in a rapidly evolving technological landscape. Additionally, the MIT program supports SDG 8: Decent Work and Economic Growth by preparing students for high-demand roles in IT, such as software development, cybersecurity, and data analytics. This focus on innovation and technological expertise not only enhances

economic productivity. The program also contributes to SDG 9: Industry, Innovation, and Infrastructure by emphasizing the development of sustainable and resilient IT infrastructure, including cloud computing, IoT, and cybersecurity solutions. Through research and innovation, students are encouraged to design technologies that address real-world challenges, such as improving access to digital services and enhancing infrastructure resilience. Furthermore, the MIT program aligns with SDG 11: Sustainable Cities and Communities by incorporating coursework on smart cities, urban planning technologies, and data-driven decision-making. Graduates will be equipped to develop IT solutions that promote sustainable urbanization, such as intelligent transportation systems, energy-efficient buildings, and digital governance platforms.

Therefore, people who possess a relatively advanced level of knowledge on information technology will have a better ability to understand and cope with the rapid changes brought about by the revolution, greater capacity to achieve the most, and more capability to contribute to society, which places M.I.T. degree holders at the top of the pack.

Thus, this is to support the university's mission of its quest to quality education and commitment to deliver instruction and services for global competitiveness and satisfaction of clientele. The College of Engineering and Computational Sciences, Department of Computational Sciences proposed offering MIT to ParSU.

## IV. PROGRAM COMPLIANCE

The Bachelor of Science in Information Technology (BSIT) program at Partido State University has consistently demonstrated its commitment to excellence and adherence to stringent educational standards. Authorized to operate under Board of Regents Resolution No. 05, series of 2003, and subsequently revised through Resolution No. 11, series of 2013, the program has been offered with a strong compliance record.

Adhering to the Policies, Standards, and Guidelines (PSG) stipulated in Commission on Higher Education Memorandum Order No. 25, series of 2015, the program has also been awarded a Certificate of Program Compliance (COPC No. 01, s. 2022) on January 17, 2022, by the Commission on Higher Education. This certification confirms the program's alignment with national educational quality standards and operational guidelines. Moreover, the BSIT program has attained a Level III Accreditation Status from the Accrediting Agency of Chartered Colleges and Universities in the Philippines (AACCUP). This top-tier accreditation status recognizes the program's excellence in curriculum implementation, faculty qualifications, research initiatives, and community involvement, positioning it among the elite IT education providers in the region.

Based on these merits, it is evident that Partido State University possesses the established infrastructure, faculty expertise, and programmatic excellence required to propose and offer a Master of Information Technology (MIT) program. The university's proven track record and its adherence to high standards of quality assurance make it exceptionally well-placed to expand its offerings and introduce an advanced degree program that will further contribute to the field of information technology.

## V. OBJECTIVE

To adopt the Master of Information Technology – Professional Track curriculum in accordance with CMO No. 15, s. 2019.

## VI. FEATURES OF THE PROPOSED CURRICULUM

The Master of Information Technology program is a two-year graduate course designed to prepare students as organizational and societal leaders who drive change by developing and implementing technology-based solutions. Throughout the program, students gain comprehensive understanding of

various organizational structures—including businesses and government entities—and acquire advanced technical skills. This training empowers them to leverage the transformative potential of ICT and effectively address strategic organizational challenges. The curriculum bridges theory and practice, encouraging students to explore relevant theories and apply them to practical challenges. This approach enables them to devise impactful solutions to complex societal issues. The program prepares candidates to navigate the multifaceted and often unpredictable nature of IT, fostering critical thinking skills that can handle ambiguous and complex situations. Additionally, students are equipped to consider and manage the ethical implications inherent in IT-related decisions.

Pursuant to Commission on Higher Education (CHED) Memorandum Order No. 15, series of 2019, the Master of Information Technology (MIT) program is a non-thesis track graduate program, designed to cultivate advanced skills and knowledge in technology and its application. The program requires students to complete 36 credit units: 12 units of core coursework that lay the foundation of theoretical and practical knowledge, 15 units of specialized major courses, 3 units of cognate courses to foster an interdisciplinary perspective, and 6 units focused on the capstone project. This capstone project, vital to the MIT curriculum, involves the submission of an independent and original research work, its successful defense, and subsequent publication in a reputable refereed scientific journal or a Scopus-indexed conference.

In addition to these academic requirements, students must pass a comprehensive examination—unless they opt for an early research publication exemption—fulfill the residency requirements and meet other university stipulations to graduate. Furthermore, a second research publication in a refereed journal is required before program completion. This structured and rigorous approach not only aligns with the educational standards set by CHED in CMO No. 15, s. 2019, but also fosters a research-oriented mindset among graduates, equipping them with the expertise necessary to lead, innovate, and contribute significantly to both industry and society.

## PROGRAM DESCRIPTION

The Master of Information Technology - Professional program prepares students for leadership roles in information technology by providing them with the necessary expertise and knowledge. The program establishes a strong foundation in information technology, allowing students to develop highly specialized, marketable skills in the areas of information technology that are most relevant to their careers and professional interests, as well as the needs of society.

## PROGRAM OBJECTIVES

- Participating in professional development or postgraduate study in order to collaborate in diverse team environments and work effectively in the IT field to make a positive contribution to society through leadership positions and/or pursuing a research career.
- Integrate and apply IT-based solutions effectively in their practice to solve problems and meet the desired needs of society.
- Demonstrate a thorough understanding of the broad concepts of information technology while exercising professionalism and ethical judgment in the development, implementation, and management of information technology systems.

## PROGRAM OUTCOMES

Graduates of the Master of Information Technology program will possess advanced knowledge and practical expertise in the field of information technology, equipping them for leadership roles in diverse professional settings. The MIT Program at Partido State University aligns with **Level 7 of the Philippine Qualifications Framework (PQF)** as mandated in **CMO No. 15, s. 2019**, which specifies that graduate programs must demonstrate:

### ADVANCED KNOWLEDGE AND SKILLS IN INFORMATION TECHNOLOGY

**PO1.** Apply advanced IT concepts, frameworks, and methodologies to design innovative, secure, and scalable technological solutions that address complex organizational and societal problems.

**PO2.** Demonstrate mastery of specialized IT domains—including systems design, data analytics, cybersecurity, and IT management—through interdisciplinary and industry-aligned approaches.

### SELF-DIRECTED RESEARCH

**PO3.** Independently conceptualize, design, and conduct high-quality IT research or capstone projects that contribute to innovation, digital transformation, and knowledge creation.

**PO4.** Critically evaluate existing IT literature, technologies, and practices to propose improvements, emerging trends, or novel approaches.

### LIFELONG LEARNING WITH A HIGH DEGREE OF INDEPENDENCE

**PO5.** Demonstrate a high degree of independence, adaptability, and reflective practice in managing continuous professional development aligned with global IT advancements.

**PO6.** Collaborate effectively with interdisciplinary or multidisciplinary teams in designing, evaluating, and delivering IT solutions.

### APPLICATION OF ADVANCED KNOWLEDGE AND SKILLS IN RESEARCH, PROFESSIONAL PRACTICE, OR CREATIVE WORK.

**PO7.** Translate advanced IT knowledge and research into effective strategies, solutions, technologies, and digital innovations that support organizational goals and societal development.

**PO8**. Lead and implement IT projects and innovations that align technological, ethical, and societal considerations.

**PO9**. Apply advanced IT skills in professional and creative work, producing original, innovative, and impactful IT solutions.

## PERFORMANCE INDICATORS

The Master of Information Technology (MIT) program at Partido State University is designed to equip students with advanced knowledge and specialized skills in information technology, preparing them for leadership roles in industry, academia, and research. Aligned with CHED Memorandum Order No. 15, s. 2019, the program integrates theoretical foundations with practical applications, ensuring graduates can address complex technological challenges in the rapidly evolving digital landscape.

To assess the effectiveness of the program and measure student competencies, Performance Indicators have been developed for each Learning and Program Outcome. These indicators serve as concrete benchmarks for evaluating students' proficiency in key IT domains, including software development, cybersecurity, data analytics, cloud computing, and IT project management. By meeting these indicators, graduates will demonstrate their ability to innovate, conduct research, lead IT initiatives, and **contribute to the advancement of technology in both organizational and societal contexts.**

| Learning Outcomes | Performance Indicators |
|---|---|
| **ADVANCED KNOWLEDGE AND SKILLS IN INFORMATION TECHNOLOGY** | |
| **PO1.** Apply advanced IT concepts, frameworks, and methodologies to design innovative, secure, and scalable technological solutions that address complex organizational and societal problems. | **a.1** Design and implement enterprise-level IT architectures integrating emerging technologies (AI, IoT, cloud, automation).<br>**a.2** Develop and optimize large-scale databases and data-driven applications supporting analytics and decision-making.<br>**a.3** Apply advanced software engineering practices (microservices, DevOps, CI/CD) to build scalable systems.<br>**a.4** Prototype innovative IT products or services that address societal priorities such as digital transformation, e-governance, smart cities, and sustainability.<br>**a.5** Apply cybersecurity principles and governance frameworks to protect digital assets and ensure system resilience. |
| **PO2.** Demonstrate mastery of specialized IT domains—including systems design, data analytics, cybersecurity, and IT management—through interdisciplinary and industry-aligned approaches. | **b.1** Conduct advanced data analytics using statistical and algorithmic techniques to generate actionable insights.<br>**b.2** Evaluate and implement secure network and cloud infrastructure solutions.<br>**b.3** Create enterprise information systems aligned with strategic organizational requirements.<br>**b.4** Integrate principles from management, computing, and design to build user-centered IT solutions. |
| **DIRECTED RESEARCH** | |
| **PO3.** Independently conceptualize, design, and conduct high-quality IT research or capstone projects that contribute to innovation, digital transformation, and knowledge creation. | **c.1** Formulate research problems addressing gaps in IT practice or theory.<br>**c.2** Apply quantitative, qualitative, mixed-methods, or design science research approaches aligned with IT standards.<br>**c.3** Produce publishable research output compliant with ethical standards, data protection laws (e.g., Data Privacy Act), and professional guidelines.<br>**c.4** Develop research-based IT solutions validated through experimentation, prototyping, simulation, or usability testing. |

| | |
|---|---|
| **PO4.** Critically evaluate existing IT literature, technologies, and practices to propose improvements, emerging trends, or novel approaches. | **d.1** Conduct systematic analysis of current IT research to identify trends, limitations, and opportunities.<br>**d.2** Apply critical and analytical thinking to evaluate technological solutions and methodologies.<br>**d.3** Present scholarly and professional arguments supported by evidence-based reasoning. |
| **LIFELONG LEARNING WITH A HIGH DEGREE OF INDEPENDENCE** | |
| **PO5.** Demonstrate a high degree of independence, adaptability, and reflective practice in managing continuous professional development aligned with global IT advancements. | **e.1** Obtain IT certifications (AWS, CISSP, Azure, PMP, Google Data Engineer, etc.).<br>**e.2** Engage in industry conferences, research colloquia, hackathons, and professional development programs.<br>**e.3** Maintain a professional portfolio showcasing IT expertise, research output, and innovation projects.<br>**e.4** Demonstrate advanced self-directed learning skills in rapidly evolving IT domains. |
| **PO6.** Collaborate effectively with interdisciplinary or multidisciplinary teams in designing, evaluating, and delivering IT solutions. | **f.1** Lead and participate in cross-functional and multicultural IT teams.<br>**f.2** Utilize collaborative development tools (Git, Jira, Miro, Azure DevOps) effectively.<br>**f.3** Demonstrate strong communication, documentation, and leadership skills in team-based IT projects.<br>**f.4** Facilitate knowledge sharing and peer mentoring in IT research or development teams. |
| **APPLICATION OF ADVANCED KNOWLEDGE AND SKILLS IN RESEARCH, PROFESSIONAL PRACTICE, OR CREATIVE WORK** | |
| **PO7.** Translate advanced IT knowledge and research into effective strategies, solutions, technologies, and digital innovations that support organizational goals and societal development. | **g.1** Produce IT solutions that enhance business processes, efficiency, automation, governance, and public service delivery.<br>**g.2** Develop prototypes aligned with SDGs (e.g., smart environment monitoring, digital governance, sustainable cities).<br>**g.3** Apply project management methodologies (Agile, Scrum, PMBOK) to ensure successful IT project delivery. |

| | |
|---|---|
| **PO8**. Lead and implement IT projects and innovations that align technological, ethical, and societal considerations. | **h.1** Integrate ethical principles and data governance frameworks into IT designs and decision-making.<br>**h.2** Develop user-centered systems addressing accessibility, inclusiveness, and societal needs.<br>**h.3** Advocate responsible, secure, and sustainable IT practices. |
| **PO9.** Apply advanced IT skills in professional and creative work, producing original, innovative, and impactful IT solutions. | **i.1** Create, deploy, and evaluate intelligent systems, digital platforms, or automated solutions that solve real-world problems.<br>**i.2** Prepare technical documentation, research manuscripts, and professional reports meeting global academic and industry standards.<br>**i.3** Present and defend IT solutions to technical panels, stakeholders, and industry partners. |

## CANDIDATE STUDENTS

The program targets individuals with a strong foundation in IT, preferably holding degrees in Information Technology, Computer Science, or related fields. Ideal candidates include professionals in IT roles such as software developers, system analysts, IT project managers, and network administrators. Leadership potential and collaboration skills are essential, as the program prepares students for multidisciplinary projects and leadership roles in industry or academia. The program is expected to attract professionals from the region, including alumni of Partido State University and other local and government institutions, as well as industry practitioners seeking to advance their expertise and careers.

## BRIDGING COURSES

Bridging courses are designed and assigned during the midyear term to address specific preparatory needs for candidate students who are non-IT or non-allied students who may not possess the required foundational knowledge for advanced coursework in the Master of Information Technology (MIT) program. These courses—*Object-Oriented Programming, Data Structures, and Trends in IT*—aim to ensure that incoming students, particularly those from non-IT backgrounds or with limited prior exposure, are equipped with the necessary competencies to succeed in the program's core and advanced subjects.

The midyear timing allows students to complete these courses before the start of the formal academic year, creating a seamless transition into the graduate program while preventing delays in their academic progress. This approach reflects the institution's commitment to inclusivity, academic readiness, and maximizing student success.

## MODE OF DELIVERY

The proposed Master of Information Technology (MIT) program will adopt a hybrid mode of delivery, integrating the following formats to cater to diverse learner needs and ensure flexibility while maintaining quality education:

- **Full Onsite Learning**:
  Scheduled in-person sessions will be conducted on campus, providing opportunities for direct interaction between students and instructors. These sessions will focus on laboratory exercises, hands-on activities, collaborative group work, and discussions that require immediate feedback and engagement.

- **Full-Online Learning:**
  Real-time virtual classes will be held using digital platforms. These sessions will allow for live interaction, lectures, discussions, and Q&A segments, simulating a traditional classroom environment in an online setting.

- **Blended Online Learning**:
  Students will have access to recorded lectures, reading materials, discussion boards, and other online resources. This self-paced approach supports flexibility, allowing learners to engage with the material at their convenience while meeting prescribed deadlines.

This hybrid approach ensures that students benefit from both the rich, interactive learning experiences of face-to-face instruction and the flexibility and accessibility of online learning modalities. It also aligns with the demands of a modern, technologically driven educational framework, suitable for preparing graduates to meet the challenges of the 4th Industrial Revolution.

# Proposed Program of Study for the Master of Information Technology - Professional Track Based on CMO 15, series of 2019

| MIDYEAR* | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| **MITB01** | Object-Oriented Programming | (3) | 3 | 0 | 3 | **None** |
| **MITB02** | Data Structure | (3) | 3 | 0 | 3 | **None** |
| **MITB03** | Trends in IT | (3) | 3 | 0 | 3 | **None** |
| **TOTAL** | | **(9)** | **9** | **0** | **0** | **None** |

* Midyear is required for non-IT/non-allied applicants

| FIRST YEAR, First Semester | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| **MIT101** | Advanced Database Systems | 3 | 3 | 0 | 3 | **None** |
| **MIT102** | Advanced Operating Systems and Networking | 3 | 3 | 0 | 3 | **None** |
| **MIT103** | Advanced System Design and Implementation | 3 | 3 | 0 | 3 | **None** |
| **MIT104** | IT Research Methods (Methods of Research) | 3 | 3 | 0 | 3 | **None** |
| **TOTAL** | | **12** | **12** | **0** | **12** | |

| FIRST YEAR, Second Semester | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| **MIT201** | Technology and Project Management | 3 | 3 | 0 | 3 | **None** |
| **MIT202** | Web Engineering | 3 | 3 | 0 | 3 | **None** |
| **MIT203** | ICT Information Security Administration | 3 | 3 | 0 | 3 | **None** |
| **MIT204** | Management Information Systems | 3 | 3 | 0 | 3 | **None** |
| **TOTAL** | | **12** | **12** | **0** | **12** | |

| MID YEAR | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| **MIT301** | Human-Computer Interaction | 3 | 3 | 0 | 3 | **None** |
| **MIT302** | Introduction to Data Analytics | 3 | 3 | 0 | 3 | **None** |
| | Comprehensive Examination | - | - | - | - | 100% Taken and passed subjects |
| **TOTAL** | | **3** | **3** | **0** | **3** | |

| SECOND YEAR, First Semester | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| MIT401 | Graduate Project 1 | 3 | 3 | 0 | 3 | Comprehensive Examination |
| **TOTAL** | | **3** | **3** | **0** | **3** | |

| SECOND YEAR, Second Semester | | | | | | |
|---|---|---|---|---|---|---|
| **Code** | **Course Title** | **Units** | **Lec** | **Lab** | **CH** | **Pre-requisite(s)** |
| MIT402 | Graduate Project 2 | 3 | 3 | 0 | 3 | MIT 401 |
| **TOTAL** | | **3** | **3** | **0** | **3** | |

## SUMMARY

| **CMO NO.15 S.2019** | |
|---|---|
| Coursework | 27 |
| Comprehensive Exam | - |
| Capstone Course | 3 |
| **Total** | **30** |

| **PROPOSED** | |
|---|---|
| Bridging Courses | (9) |
| Core Courses | 12 |
| Major Courses | 12 |
| Cognate Course | 6 |
| Comprehensive Exam | - |
| Graduate Project (Capstone) | 6 |
| **Total** | **36** |

## VII. CURRICULUM COMPONENT

The MIT curriculum is based on the Commission on Higher Education's (CHED) Policies and Standards for Graduate Programs in Information Technology Education and is benchmarked against the curriculum of leading international academic institutions offering this program.

**Program of Study**

### Bridging Courses

**MITB01 Object-Oriented Programming**
**MITB02 Data Structure**
**MITB03 Trends in IT**

### Core Subjects (12 units)

**MIT101 Advanced Database Systems**
**MIT102 Advanced Operating Systems and Networking**
**MIT103 Advanced System Design and Implementation**
**MIT104 IT Research Methods (Methods of Research & Seminar on Thesis Writing)**

### Major Subjects (12 units)

**MIT 201 Technology and Project Management**
**MIT 202 Web Engineering**
**MIT 203 ICT Information Security Administration**
**MIT 204 Management Information Systems**

### Cognates (6 units)

**MIT 301 Human-Computer Interaction**
**MIT 302 Introduction to Data Analytics**

### Comprehensive Examination

### Graduate Project (6 units)

**MIT 401 Graduate Project 1**
**MIT 402 Graduate Project 2**

**TOTAL NUMBER OF UNITS: 36**

**MIT Course Descriptions**

**BRIDGING COURSES**

| Course Name | Object-Oriented Programming |
|---|---|
| Course Code | MITB01 |
| Course Description | This course provides a comprehensive study of Object-Oriented Programming (OOP) principles, focusing on the design, development, and implementation of object-oriented applications. Students will learn fundamental concepts such as encapsulation, inheritance, polymorphism, and abstraction using an industry-relevant programming language (e.g., Java, C++, Python). The course emphasizes software design patterns, reusable code structures, and best practices in software engineering. Hands-on activities, case studies, and a final project will provide students with the ability to apply OOP methodologies to real-world applications |
| Course Credit | 3 units |
| Contact Hours | 3 Lecture Hours / Week |
| Prerequisite | None |

| Course Name | Data Structures |
|---|---|
| Course Code | MITB02 |
| Course Description | This course explores fundamental and advanced data structures essential for efficient algorithm design and software development. Topics include arrays, linked lists, stacks, queues, trees, graphs, hash tables, heaps, and their associated algorithms. The course emphasizes memory management, algorithmic efficiency (Big-O notation), and best practices in data manipulation. Through hands-on programming exercises, case studies, and a final project, students will develop the ability to design, implement, and optimize data structures in real-world applications |
| Course Credit | 3 units |
| Contact Hours | 3 Lecture Hours / Week |
| Prerequisite | None |

| Course Name | Trends in Information Technology |
|---|---|
| Course Code | MITB03 |
| Course Description | This course explores the latest trends and emerging technologies in Information Technology (IT), focusing on innovations that drive industry and society. Topics include cloud computing, artificial intelligence, big data analytics, cybersecurity advancements, blockchain technology, quantum computing, edge computing, and the impact of IT on business and society. The course will examine ethical considerations, regulatory challenges, and IT's role in achieving sustainability. Through case studies, research, and hands-on exploration, students will gain insights into IT's evolving landscape and its future directions. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

**CORE COURSES**

| Course Name | Advanced Database Systems |
|---|---|
| Course Code | MIT101 |
| Course Description | This course offers a comprehensive study of the design, implementation, and management of modern database systems. It covers core principles and techniques in relational databases, query optimization, and data and information management, while exploring advanced topics such as distributed databases, NoSQL systems, data warehousing, and big data management. Students will gain hands-on experience with cutting-edge database technologies and tools, enabling them to design scalable, efficient, and secure database solutions. The course also examines emerging trends, including cloud-based databases, real-time data processing, and future trends in database design, equipping students with the skills to address the challenges of managing large-scale data in today's data-driven world. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | Advanced Operating Systems and Networking |
|---|---|
| Course Code | MIT102 |
| Course Description | This course provides an in-depth understanding of the fundamental concepts of modern operating systems and computer networks. It explores advanced topics in operating systems, including parallel processing systems, distributed systems, real-time systems, network operating systems, open-source operating systems, and cloud computing. Students will gain a comprehensive understanding of how these systems are designed, implemented, and managed, with a focus on their role in supporting complex computing environments. The course also covers the integration of operating systems with networking technologies, enabling students to analyze and optimize system performance in distributed and cloud-based environments. Through theoretical knowledge and practical applications, students will develop the skills needed to design, manage, and troubleshoot advanced operating systems and networked computing systems. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | Advanced System Design and Implementation |
|---|---|
| Course Code | MIT103 |
| Course Description | This course focuses on the systematic analysis, design, and implementation of complex information systems. Students will explore comparative development methodologies and modeling tools to understand the end-to-end process of system development. Key topics include the design of human-computer interfaces, system implementation strategies, and system maintenance practices. The course emphasizes the application of theoretical concepts to real-world scenarios, enabling students to develop, deploy, and maintain robust information systems. Through hands-on projects and case studies, students will gain practical experience in using advanced tools and techniques to design user-friendly, efficient, and scalable systems. By the end of the course, students will be equipped with the skills to successfully manage the lifecycle of information systems, from initial analysis to post-implementation maintenance. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | IT Research Methods |
|---|---|
| Course Code | MIT 104 |
| Course Description | This course provides students with a comprehensive introduction to research methodologies within the field of information technology. It covers a broad range of research-related topics, including major research paradigms, research design principles, research ethics, and data collection and analysis techniques. Students will explore various research methods applicable to IT and its sub-disciplines, such as quantitative, qualitative, and mixed-method approaches. The course emphasizes the development of critical thinking and problem-solving skills, enabling students to design, conduct, and evaluate research projects effectively. By the end of the course, students will be equipped with the knowledge and tools to undertake rigorous IT research, contributing to advancements in the field while adhering to ethical standards and best practices. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

**MAJOR COURSES**

| Course Name | Technology and Project Management |
|---|---|
| Course Code | MIT201 |
| Course Description | This course explores technology and project management from a managerial perspective, focusing on the essential processes and techniques required to successfully plan, execute, and oversee projects. Students will be introduced to the fundamental principles of project management, including project planning, resource allocation, risk management, and project evaluation. The course also covers technology management techniques and behavioral skills necessary for leading diverse teams and managing complex projects, particularly in the context of instructional design and IT initiatives. Through case studies and practical exercises, students will learn to apply systematic approaches to project management, ensuring the successful delivery of projects on time, within scope, and on budget. By the end of the course, students will be equipped with the tools and strategies to effectively manage technology-driven projects in various organizational settings. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | Human Computer Interaction |
|---|---|
| Course Code | MIT202 |
| Course Description | This course introduces students to the interdisciplinary field of Human-Computer Interaction (HCI), which integrates principles from computer science, cognitive psychology, design, and other related disciplines. Students will explore the fundamental aspects of human perception, cognition, and learning as they apply to the design, implementation, and evaluation of user interfaces. The course emphasizes both theoretical knowledge and hands-on experience, enabling students to create intuitive, user-friendly, and accessible interfaces. Topics include usability testing, user-centered design, interaction design, and evaluation techniques. Through practical projects and case studies, students will develop the skills to design and evaluate effective interfaces that enhance user experience and meet the needs of diverse user groups. By the end of the course, students will be equipped to apply HCI principles to real-world challenges in technology design and development. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | ICT Information Security Administration |
|---|---|
| Course Code | MIT203 |
| Course Description | This course provides a comprehensive overview of data security, integrity, privacy, and administration in the context of information and communication technology (ICT). Students will explore the principles of computer security, including the development and application of security policies, mechanisms for policy enforcement, and technologies for vulnerability and intrusion detection. The course covers key topics such as risk management, access control, encryption, network security, and incident response. Through case studies and practical exercises, students will learn to identify security threats, implement protective measures, and manage security systems effectively. By the end of the course, students will be equipped with the knowledge and skills to administer and safeguard ICT systems, ensuring data protection and compliance with security standards in organizational environments. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | Management Information Systems |
|---|---|
| Course Code | MIT204 |
| Course Description | This course focuses on the strategic role of information technology (IT) in modern organizations, teaching students to evaluate the opportunities and challenges managers face when leveraging IT to add value to their organizations. Students will explore how management information systems (MIS) support decision-making, operational efficiency, and competitive advantage. The course also examines the transformational changes driven by IT within and across industries, including the impact of emerging technologies such as big data, cloud computing, and artificial intelligence. Through case studies and real-world examples, students will develop the skills to analyze, design, and implement IT solutions that align with organizational goals. By the end of the course, students will be equipped to understand and manage the dynamic interplay between technology, business processes, and organizational strategy. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

| Course Name | Web Engineering |
|---|---|
| Course Code | MIT205 |
| Course Description | This course introduces a structured methodology for Web development projects, drawing parallels to principles used in software engineering. Students will learn the fundamental concepts, methods, technologies, and techniques essential for designing, developing, and maintaining effective websites and web applications. The course covers key topics such as web architecture, information organization, user interface design, database integration, and web security. Emphasis is placed on creating websites that efficiently collect, organize, and expose information resources while ensuring usability, scalability, and maintainability. Through hands-on projects and case studies, students will gain practical experience in applying web engineering principles to real-world scenarios. By the end of the course, students will be equipped with the skills to systematically plan, develop, and manage web-based systems that meet user and organizational needs. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

**COGNATE COURSES**

| Course Name | Introduction to Data Analytics |
|---|---|
| Course Code | MIT301 |
| Course Description | This course provides a comprehensive introduction to data analytics and visualization, equipping students with the skills to extract, process, analyze, and interpret data effectively. Students will learn how to access data from diverse sources, including web APIs and repository servers, and how to clean and organize data for analysis. The course covers the application of analytical techniques to solve real-world problems and emphasizes the creation of data visualizations to communicate insights clearly. Hands-on instruction will be provided in relevant programming languages (e.g., Python, R) and popular data analytics and visualization tools (e.g., Tableau, Power BI). By the end of the course, students will be able to perform end-to-end data analysis, from data acquisition to visualization, and apply these skills to practical, data-driven decision-making scenarios. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **None** |

**GRADUATE PROJECT**

| Course Name | Graduate Project 1 |
|---|---|
| Course Code | MIT401 |
| Course Description | This is the first course in a two-part sequence. Students complete unique capstone projects that synthesize and integrate concepts and principles taught throughout the MIT program. This course applies IT project management practices by having students define and design projects in the areas of information technology management, such as database management, networking, and security. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **Comprehensive Exam** |
| **Course Name** | **Graduate Project 2** |
| Course Code | MIT 402 |
| Course Description | This course is a continuation of MIT 401, in which students complete the project defined and designed in MIT 401. Students should implement, code, test, and document their projects prior to actually presenting them in front of a review panel for final evaluation and approval. |
| Course Credit | **3 units** |
| Contact Hours | **3 Lecture Hours / Week** |
| Prerequisite | **MIT401** |

### Curriculum Map

Program Outcomes (POs): PO1 Advanced IT solutions; PO2 Specialized domain mastery; PO3 Self-directed research; PO4 Critical evaluation; PO5 Lifelong learning; PO6 Teamwork/leadership; PO7 Organizational/societal application; PO8 Ethics/security/accessibility; PO9 Professional/creative IT work.

| Course Code | Course Title | Units | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| MITB01 | Object-Oriented Programming | 3 | I | I | | | | | | | |
| MITB02 | Data Structure | 3 | I | I | | I | | | | | |
| MITB03 | Trends in IT | 3 | | I | | | I | | I | | |
| MIT101 | Advanced Database Systems | 3 | I/R | R | | I | | | | I | R |
| MIT102 | Advanced Operating Systems & Networking | 3 | R | R | | | | | I | R | I |
| MIT103 | Advanced System Design & Implementation | 3 | R | R | | I | | R | I | I | R |
| MIT104 | IT Research Methods (Methods of Research) | 3 | | | I/R | R | | | | R | |
| MIT201 | Technology & Project Management | 3 | I | I | | | R | R | R | | |
| MIT202 | Human-Computer Interaction | 3 | I | R | | I | | | I | R | R |
| MIT203 | ICT Information Security Administration | 3 | R | R | | | | | I | M | I |
| MIT204 | Management Information Systems | 3 | I | R | | R | | | R | I | |
| MIT205 | Web Engineering | 3 | R | R | | | | I | R | I | R |
| MIT301 | Introduction to Data Analytics | 3 | I | R | | | | | R | | I |
| MIT401 | Graduate Project 1 | 3 | R | R | R | R | I | R | R | R | R |
| MIT402 | Graduate Project 2 | 3 | M | M | M | M | R | M | M | M | M |

Legend: I = Introduced, R = Reinforced/Developed, M = Mastery/Assessed

## VIII. Faculty

Since the proposed program is under the Non-thesis track, CMO No. 19, 2019 requires the following number of qualified faculty for the graduate program.

| CMO No. 19, s. 2019 | Proposed MIT Program Faculty Component |
|---|---|
| **a. For Master's Programs, there shall be at least four (4) faculty members per program at all times.**<br>• For the non-thesis track<br>○ Four (4) full time faculty who have doctoral degrees in the discipline and with published works in refereed journal/s and/or have produced publicly-recognized creative and/or technology outputs; or<br>○ If there is a dearth in doctoral degree holders in the discipline, at least one (1) full-time faculty who has a doctoral degree in the discipline and with published works in refereed journal/s and/or have produced publicly-recognized creative and/or technology outputs; and at least:<br>▪ *three (3) full-time faculty who are master's degree holders, but with doctoral units in the* | In alignment with CMO No. 19, s. 2019, the Master of Information Technology (MIT) program's faculty component has been strategically designed to ensure the delivery of a high-quality, industry-relevant graduate education. The proposed faculty roster meets and exceeds the requirements specified for non-thesis track programs, addressing both academic qualifications and professional experience as follows:<br>1. **Doctoral Degree Holders in the Discipline**:<br>a. The program includes three (3) full-time faculty members who hold doctoral degrees in Information Technology. These scholars bring significant expertise and research proficiency to the program, with numerous publications in refereed journals, showcasing their academic leadership in the field.<br>2. **Master's Degree Holders with Doctoral Units and Publications**:<br>a. The roster incorporates three (3) full-time faculty members who possess master's degrees in IT and related fields, have completed doctoral units, and |

| | |
|---|---|
| *discipline and have at least one (1) publication in a refereed journal. However, they can only teach courses, they cannot serve as thesis advisers; or*<br>▪ *for industry-oriented programs, three (3) full-time faculty who are master's degree holders, but with extensive industry experience. However, they can only teach courses, they cannot serve as thesis/capstone advisers* | have at least one (1) publication in referred journals. While these faculty members focus on teaching, they enrich the program with advanced academic perspectives and up-to-date knowledge.<br>b. These faculty members also serve as exemplars of continuous professional development, encouraging a culture of lifelong learning among students.<br>3. **Industry-Oriented Faculty**:<br>a. To further enhance the program's practical orientation, the faculty includes professionals with extensive industry experience in systems development, web design, and management. This ensures that students benefit from real-world insights and hands-on expertise.<br>4. **Diverse Specializations**:<br>a. The faculty's diverse expertise spans across critical areas such as systems development, graphics and computer vision, management, and web development. This multidisciplinary approach enriches the program, fostering a holistic and adaptable learning environment.<br>5. **Strengthened by Emerging Scholars**:<br>a. Several faculty members are currently pursuing doctoral studies, ensuring a pipeline of emerging scholars who will contribute to the program's sustainability and continuous improvement.<br>6. **Commitment to Excellence**:<br>a. All faculty members demonstrate a commitment to both academic and professional excellence, evidenced by their credentials, publications, and contributions to the field. This ensures that the MIT program remains competitive, relevant, and in tune with the demands of the Fourth Industrial Revolution.<br><br>The proposed faculty composition fulfills the requirements of CMO No. 19, s. 2019 and positions the MIT program as a premier graduate offering, fostering advanced skills and leadership capabilities in Information Technology. |

**Doctorate Degree holders**

| | | | |
|---|---|---|---|
| 1 |  | **ATOLE, Ronnel Rimando**<br>Professor VI | 13 papers |
| | | **Bachelor of Law**, University of Northeastern Philippines, Iriga City, Philippines — Law | |
| | | **PhD in Information Technology**, Hannam University, South Korea — Information Technology | |
| | | **Master of Science in Information Technology**, Hannam University, South Korea — Information Technology | |
| | | **Bachelor of Science in Computer Science**, Ateneo de Naga University, Naga City, Philippines — Computer Science | |
| 2 |  | **CAPUCAO, Joni Neil Borja**<br>Associate Professor V | 5 |
| | | **Doctor in Information Technology**, University of the Cordilleras, Baguio City, Philippines — Information Technology | |
| | | **Master of Information Technology,** Ateneo de Naga University, Naga City, Philippines — Multimedia and Object-Oriented Technology Applications | |
| | | **Bachelor of Science in Mathematics,** Ateneo de Naga University, Naga City, Philippines — Mathematics | |
| 3 |  | **CUYA, Kennedy Centeno**<br>Assistant Professor II | 4 |
| | | **Doctor in Information Technology**, University of the Cordilleras, Baguio City, Philippines — Information Technology | |
| | | **Master of Information Technology,** Ateneo de Naga University, Naga City, Philippines — Systems Development | |
| | | **Bachelor of Science in Information Technology,** Partido State University, Camarines Sur, Philippines — Information Technology | |

**Master Degree holders**

| | | FACULTY | Field of Specialization | Published paper |
|---|---|---|---|---|
| 1 |  | **ARTIAGA, Rowell John Baterna**<br>Assistant Professor II<br>**Doctor in Information Technology (CAR)**, University of the Cordilleras, Baguio City, Philippines<br>**Master of Information Technology,** Ateneo de Naga University, Naga City, Philippines<br>**Bachelor of Science in Information Technology,** Partido State University, Camarines Sur, Philippines | <br><br>Information Technology<br>Systems Development<br>Information Technology | 10 |
| 2 |  | **BRIONES, Salvador II V.**<br>Associate Professor<br>**Doctor in Information Technology (CAR)**, AMA University and Colleges, Quezon City, Philippines<br>**Master of Science in Web Design and Development**, UNICAF University, Zambia<br>**Master of Arts in Education major in Mathematics,** Camarines Sur, Philippines<br>**Bachelor of Science in Information Technology,** Partido State University, Camarines Sur, Philippines | <br><br>Information Technology<br>Web Design and Development<br>Mathematics Education<br>Information Technology | 11 |
| 3 |  | **BRIONES, Shane C.**<br>Assistant Professor II<br>**Doctor in Information Technology (with Units)**, AMA University and Colleges, Quezon City, Philippines<br>**Master of Science in Web Design and Development**, UNICAF University, Zambia<br>**Master of Arts in Education major in Mathematics,** Camarines Sur, Philippines<br>**Bachelor of Science in Computer Science,** Ateneo de Naga University, Naga City, Philippines | <br><br>Information Technology<br>Web Design and Development<br>Mathematics Education<br>Computer Science | 5 |
| 4 |  | **ABIO, Arjay Formalejo**<br>**Assistant Professor II**<br>**Master of Information Technology,** Ateneo de Naga University, Naga City, Philippines<br>**Bachelor of Science in Information Technology,** Partido State University, Camarines Sur, Philippines | <br><br>Systems Development<br>Information Technology | 1 |

| | | | | |
|---|---|---|---|---|
| 5 |  | **PAGA, Jonathan**<br>Instructor III | | 2 |
| | | **Doctor in Information Technology (with units)**, Polytechnic University of the Philippines, Manila, Philippines | Information Technology | |
| | | **Master of Information Technology,** Aemillianum College, Sorsogon, Philippines | Information Technology | |
| | | **Bachelor of Science in Computer Science,** University of Nueva Caceres, Naga City, Philippines | Computer Science | |
| 6 |  | **PESINO, Hazel Jean C.**<br>Assistant Professor III | | 3 |
| | | **PhD in Development Management (CAR)**, Bicol University, Legazpi City, Philippines | Development Management | |
| | | **Master of Information Technology,** Aemillianum College, Sorsogon, Philippines | Information Technology | |
| | | **Bachelor of Science in Computer Science,** Lower Isarog Foundation Exponent, Inc., Camarines Sur, Philippines | Computer Science | |
| 7 |  | **INDICO, Michael H**<br>Assistant Professor III | | 2 |
| | | **Master of Science in Computer Science (w/ Units)**, University of the Philippines – Diliman, Quezon City, Philippines | Computer Science | |
| | | **Master of Science in Computer Engineering**, Mapua University, Manila, Philippines | Computer Engineering | |
| | | **Master in Management**, Partido College, Camarines Sur, Philippines | Management | |
| | | **Bachelor of Science in Computer Engineering**, Aquinas University, Legazpi City, Philippines | Computer Engineering | |
| 8 |  | **BELLO, Leo Constantine S.**<br>Information Technology Officer I | | 8 |
| | | **Master of Science in Computer Science**, Ateneo de Naga University, Naga City, Philippines | Graphics and Computer Vision | |
| | | **Bachelor of Science in Information Technology**, AMA Computer College, Naga City | Information Technology | |

The proposed Master of Information Technology program is supported by distinguished faculty, ensuring that students receive an education that is not only comprehensive and at the forefront of technological advancement. The faculty roster includes ten (10) exceptional educators, each bringing an expertise and specialized knowledge to the program.

Among our esteemed faculty, three hold doctoral degrees in Information Technology, demonstrating profound insights and expertise in the field. These scholars are complemented by five additional faculty members with master's degrees in highly relevant disciplines such as Computer Science, Computer Engineering, and specialized areas of Information Technology including Graphics and Computer Vision, Systems Development, and Management. This blend of advanced qualifications ensures that our program offers a deep dive into both the theoretical and practical aspects of IT.

Further enriching our faculty's capabilities, we have members currently pursuing doctoral degrees in relevant fields, who are anticipated to complete their studies by the time the program is officially offered. This not only augments our faculty's scholarly depth but also ensures fresh, competent, and cutting-edge perspectives are continuously integrated into the curriculum.

Notably, two (2) faculty members uniquely bridges multiple disciplines, holding master's degrees in web design and development as well as Mathematics Education, further enriching our program's multidisciplinary approach. This diverse educational background among the faculty not only enhances the learning environment but also exemplifies our commitment to providing an education that is both broad in scope and specific in focus.

Our faculty are not just educators but also leaders in their fields, equipped to guide students as they navigate the complexities of Information Technology. The faculty's academic credentials, combined with their professional achievements, position them as exemplary figures capable of inspiring and mentoring the next generation of IT professionals. This dynamic team ensures that our Master of Information Technology program remains at the forefront of technological education, preparing students to excel in a rapidly evolving digital landscape.

**Prepared by:**

**JONI NEIL B. CAPUCAO, DIT**
Department Chair, Department of Computational Sciences

**KENNEDY C. CUYA, DIT**
Graduate Program Coordinator, MIT

**ROWELL JOHN B. ARTIAGA, MIT**
Associate Dean, CEC

**HAZEL JEAN C. PESINO, MIT**
Faculty, BSIT

**ENGR. EMELINA R. PADAYO**
Dean, College of Engineering and Computational Sciences